Arista Access Node HSR-ACC-27 Routing Protocol Flap SOP

Node: HSR-ACC-27 | **Vendor:** Arista | **Severity:** Warning | **Alarm Code:** ALARM_4810

Description

This Standard Operating Procedure addresses a high-severity warning failure on Arista Access Node HSR-ACC-27 located at HSR Layout Access, where the routing protocol flap is observed, characterized by BGP route churn, OSPF adjacency loss, and ISIS LSP flooding. The root cause of this failure domain lies in the misconfiguration or hardware anomaly affecting the control plane operations of the Arista Access switch, likely due to a discrepancy in routing protocol adjacencies or a hardware-related issue impacting the LSP (Label Switched Path) establishment and maintenance in the ISIS (Intermediate System to Intermediate System) domain. The routing protocol flap can be caused by a variety of factors, including, but not limited to, neighbor adjacency issues, OSPF area configuration mismatches, BGP route reflection, or ISIS LSP database corruption. A thorough understanding of the underlying routing protocol mechanics, including BGP, OSPF, and ISIS, is crucial in diagnosing and resolving this failure domain.

Pre-requisite Checks

- Verify the site location and node ID (HSR-ACC-27) to ensure the correct device is being targeted.

- Check the status of redundant links to ensure high availability.

- Confirm the network team has been dispatched to the site to perform any necessary physical checks or repairs.

- Gather relevant logs and configuration files from the Arista Access switch for further analysis.

Potential Root Causes

Cause	Probability
BGP Neighbor Adjacency Loss	High
OSPF Area Configuration Mismatch	Medium
ISIS LSP Database Corruption	Low

Diagnostic Commands & Expected Outputs

```
Command: show bgp neighbors, Expected Output: Neighbor ID: 10.10.10.10 State: Established Address: 10.10.10.10:179 Local Address: 10.10.10.27:179, Description: This command displays the BGP neighbor status, including the local and remote addresses, state, and AS number. It helps in identifying any issues with BGP neighbor adjacencies.
```

```
Command: show ospf neighbors, Expected Output: Neighbor ID:
10.10.10.10 State: Full Address: 10.10.10.10:520, Description: This
command displays the OSPF neighbor status, including the local and
remote addresses, state, and OSPF area number. It helps in identifying
any issues with OSPF neighbor adjacencies.

Command: show isis database, Expected Output: ISIS Database : Level-1
LSPs: 10, Level-2 LSPs: 5, Description: This command displays the ISIS
database, including the number of LSPs at each level. It helps in
identifying any issues with LSP establishment and maintenance.
```

Resolution Steps

Step 1: Step: Verify BGP Neighbor Adjacency, Command: show bgp neighbors, Description: Verify the BGP neighbor status and ensure it is established.

Step 2: Step: Verify OSPF Neighbor Adjacency, Command: show ospf neighbors, Description: Verify the OSPF neighbor status and ensure it is full.

Step 3: Step: Verify ISIS LSP Database, Command: show isis database, Description: Verify the ISIS database and ensure LSPs are established correctly.

Step 4: Step: Reset BGP Neighbor, Command: clear bgp neighbor 10.10.10.10, Description: Reset the BGP neighbor adjacency to ensure a fresh start.

Step 5: Step: Reset OSPF Neighbor, Command: clear ospf neighbor 10.10.10.10, Description: Reset the OSPF neighbor adjacency to ensure a fresh start.

Step 6: Step: Clear ISIS LSP Database, Command: clear isis database, Description: Clear the ISIS LSP database to ensure a fresh start.

Post-Check Verification

```
Command: show bgp neighbors, Expected Output: Neighbor ID: 10.10.10.10
State: Established Address: 10.10.10.10:179 Local Address:
10.10.10.27:179, Description: Verify the BGP neighbor status and
ensure it is established.

Command: show ospf neighbors, Expected Output: Neighbor ID:
10.10.10.10 State: Full Address: 10.10.10.10:520, Description: Verify
the OSPF neighbor status and ensure it is full.

Command: show isis database, Expected Output: ISIS Database : Level-1
LSPs: 10, Level-2 LSPs: 5, Description: Verify the ISIS database and
ensure LSPs are established correctly.
```

Roll-back Procedures

- Step: Restore Original Configuration, Command: copy startup-config running-config, Description: Restore the original configuration to ensure a known good state.

- Step: Reset BGP Neighbor, Command: clear bgp neighbor 10.10.10.10, Description: Reset the BGP neighbor adjacency to ensure a fresh start.

- Step: Reset OSPF Neighbor, Command: clear ospf neighbor 10.10.10.10, Description: Reset the OSPF neighbor adjacency to ensure a fresh start.

- Step: Clear ISIS LSP Database, Command: clear isis database, Description: Clear the ISIS LSP database to ensure a fresh start.